

Dynamic Pricing Optimization for Centric Brands

— Technical Brief

This document is a structured walk-through of how **dynamic price optimization** can be modeled and solved in a **realistic retail scenario**, with constraints that Centric Brands would care about. It is designed to demonstrate both the **business relevance** and the **mathematical rigor** you would expect in a technical interview.

1. Business Context

Centric Brands manages multiple categories (e.g. denim, hoodies, accessories) across channels. Key challenges include: - **Inventory risk**: avoid leftover stock while preserving margin. - **Channel constraints**: MAP (minimum advertised price), parity with retail partners, clearance floors. - **Stability constraints**: avoid large week-to-week price swings. - **Margin constraints**: ensure each sale covers at least a minimum markup.

The **goal** is to set a price path across the season (8–12 weeks, for example) that maximizes total profit, while respecting brand and channel constraints.

1.a Scenario Assumptions (used in the numerical example)

- **Product (primary)**: Women's Denim Jacket (owned brand).
Secondary (optional): Kids Hoodie (owned brand) — used when showing multi-SKU or cannibalization.
- **Channel**: Direct-to-Consumer (D2C) e-commerce on Centric-managed sites. (Wholesale/ marketplaces are referenced via parity constraints when included.)
- **Horizon & cadence**: **8 weeks, weekly** price decisions (can be daily in production; weekly is used here for clarity and to match seasonality/promos).
- **Final price grids (discrete)**:
 - **Denim (A)**: {\$79, \$89, \$99, \$109, \$119, \$129, \$139}
 - **Hoodie (B)**: {\$39, \$44, \$49, \$54, \$59, \$64, \$69, \$74}
- **Costs (COGS)**: Denim **\$38**; Hoodie **\$18**.
- **Floors**: Denim MAP **\$99**; Clearance floor **\$79** (Weeks **5 & 8**).
Hoodie MAP **\$49**; Clearance floor **\$39** (Weeks **5 & 8**).
- **Price-move caps**: Denim **\$10** per week; Hoodie **\$8** per week.
- **Initial inventories**: Denim **400** units; Hoodie **300** units.
- **End-of-season leftover cap**: $\leq 5\%$ of initial inventory.
- **Competitor price paths**: Included as exogenous features in demand; parity constraints can be turned on/off (off in the base example; on in the parity section).

These assumptions drive the demand curves and all DP optimization shown below.

2. Demand Modeling Choices

The core of pricing optimization is a **demand model**: how sales volume responds to price.

(a) Constant Elasticity (Log-Log)

$$D(p) = A \cdot \left(\frac{p}{\text{MSRP}}\right)^\varepsilon$$

- $\varepsilon < 0$: constant elasticity (e.g., -1.2 to -2.0). - Pros: interpretable, widely used, easy to estimate (linear regression in logs). - Cons: assumes elasticity is constant across the price band.

(b) Log-Linear (Semi-Log)

$$\ln D = \alpha + \beta p \quad \Rightarrow \quad D(p) = Ae^{\beta p}$$

- Pros: absolute price has constant impact (useful at lower price bands). - Cons: may under- or over-estimate elasticity away from calibration point.

(c) Hybrid / Piecewise

- Splines or tree-based models to capture non-linearities.
- Useful with rich transaction data, but harder to embed in optimization.

Why Log-Log here? - Simple, interpretable, scale-free. - Elasticity estimates are directly actionable ("1% price cut → 1.4% more sales"). - Good fit for apparel categories with broad price bands.

3. Constraints to Include

We embed realistic brand rules: 1. **Price grid**: discrete candidate prices (e.g., {79, 89, 99, 109, 119, 129}). 2. **MAP / clearance floors**: - Normal weeks: $p_t \geq \text{MAP}$. - Clearance weeks: $p_t \geq \text{ClearanceFloor}$. 3. **Price move cap**: $|p_t - p_{t-1}| \leq \Delta_{\max}$. 4. **Margin rule**: $p_t - c \geq m_{\min}$. 5. **Inventory flow**: $I_t = I_{t-1} - s_t$. 6. **Leftover cap**: $I_T \leq L_{\text{target}}$ (e.g., $\leq 5\%$ of initial stock).

4. Optimization Formulation

We want to maximize total profit:

$$\max_{\{p_t\}} \sum_{t=1}^T (p_t - c) \cdot s_t$$

subject to the constraints above.

Because decisions today affect inventory tomorrow, we solve it as a **dynamic program**.

Bellman Equation (core)

$$V(t, I, p_{t-1}) = \max_{p_t \in \text{Feasible}(t, p_{t-1})} \left[(p_t - c) \cdot s_t + V(t+1, I - s_t, p_t) \right],$$

with $s_t = \min\{I, \lfloor D_t(p_t) \rfloor\}$ and terminal condition

$$V(T+1, I, \cdot) = \begin{cases} 0, & I \leq L_{\text{target}} \\ -\infty, & I > L_{\text{target}} \end{cases}.$$

This recursively enforces the leftover constraint and bakes in all price feasibility rules (MAP/clearance, price-move cap, min margin, parity when enabled).

4.a Bellman in Action — Step-by-Step Numeric (mini 2-week demo)

Toy setup (Denim A) for intuition: - Horizon: **2 weeks** ($t = 1, 2$). - Start inventory ($t=1$): **I = 200**. - Cost **c = \$38**. - Price grid **{99, 109, 129}**. - Demand (log-log): $D(p) = A (p/129)^\varepsilon$ with $A = 120$, $\varepsilon = -1.4$. - End leftover cap: ≤ 10 units.

Terminal (Week 2): given I_2 , the value is $\max_p (p - c) \cdot \min(I_2, \lfloor D(p) \rfloor)$ if $I_2 \leq 10$, else infeasible.

Week 1: for each p_1 , compute profit now + best Week-2 value. - $p_1 = 129$: sales=120, profit=10,920; next inventory=80; Week-2 value=7,280 → **Total=18,200**. - $p_1 = 109$: sales=144, profit=10,224; next inventory=56; Week-2 value=5,096 → **Total=15,320**. - $p_1 = 99$: sales=165, profit=10,065; next inventory=35; Week-2 value=3,185 → **Total=13,250**.

Decision: choose **129** at Week 1 because it maximizes **profit now + future value**.

In the full 8-week DP, the same recursion applies each week, with seasonality, marketing lifts, floors, move caps, and leftover cap encoded in the terminal value.

5. Numerical Example (8-week horizon)

Scenario (Denim A): - MSRP: \$129; Cost: \$38; Inventory: 400; Elasticity -1.4. - Price grid: {79, 89, 99, 109, 119, 129, 139}. - Floors: MAP \$99, clearance \$79 (weeks 5, 8). - Move cap: \$10. - Leftover cap: 20 units.

Optimal 8-week price path (DP result): - Denim (A): **[129, 129, 119, 109, 99, 109, 109, 99]** - Hoodie (B): **[69, 69, 64, 59, 54, 54, 49, 49]**

Why this path? - **Weeks 1–2:** hold premium prices (129/69) while demand is strong and inventory not at risk. - **Weeks 3–4:** step down gradually within move caps to stimulate volume. - **Week 5 (promo lift + clearance):** drop to 99/54 to unlock volume at peak demand. - **Weeks 6–7:** stabilize prices (109/54→49) to balance margin vs. sell-through. - **Week 8 (final clearance):** finish low (99/49) to ensure leftover $\leq 5\%$ of stock.

This plan respects all constraints (floors, move caps, margins, leftover cap) and achieves high sell-through with controlled markdowns.

6. Extensions

- **Stochastic demand:** multiply $D(p)$ by a lognormal shock; use robust DP (pessimistic quantile) for downside protection.
 - **Cannibalization:** model two SKUs jointly with cross-elasticities $D_A(p_A, p_B), D_B(p_A, p_B)$.
 - **Reinforcement Learning:** replace DP with Q-learning/DQN when demand model is uncertain and you want the system to learn directly from interaction.
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7. Why This Matters for Centric Brands

- **Direct match to reality:** leftover caps, MAP rules, price-move caps, and clearance windows are exactly the constraints Centric faces.
 - **Actionable output:** the model produces a **week-by-week price path**.
 - **Flexibility:** can integrate parity with partners, stochastic demand, or RL exploration.
 - **Competitive edge:** turns pricing into a **systematic, data-driven lever**, not manual guesswork.
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8. Key Takeaways for Interview

1. **I understand the business constraints.** MAP, clearance, move caps, margin floors, inventory are all included.
2. **I can write the optimization mathematically.** Clear Bellman equation with constraints.
3. **I can do the numbers.** Step-by-step numeric example and full 8-week plan.
4. **I know alternatives.** Log-log vs log-linear vs hybrid, and why log-log is reasonable here.
5. **I see the future.** Mentioned stochastic demand, cannibalization, and reinforcement learning as extensions.

This is the kind of system Centric Brands could deploy today to unlock incremental margin while respecting brand guardrails.